

Vulnerability Assessment Report

OWASP Juice Shop

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Cyber Security Internship – Task 1

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Objective

The objective of this task is to perform a basic security assessment on a web application in order to identify potential vulnerabilities and understand their impact on system security.

- Identify common web vulnerabilities
 - Analyze their potential risks
 - Suggest mitigation strategies

Target Application

OWASP Juice Shop

This is an intentionally vulnerable web application developed for security training and educational purposes. It simulates real-world security flaws.

URL:

<https://juice-shop.herokuapp.com>

Tools Used

- Nmap

Used for network scanning and identifying open ports and services.

- OWASP ZAP

Used for automated vulnerability scanning and detecting security issues.

- Mozilla Firefox

Used for manual testing and analyzing application behavior.

Methodology

1. Reconnaissance

Explored the application structure, pages, and input fields.

2. Network Scanning

Performed basic scanning using Nmap to identify accessible services.

3. Automated Scanning

Used OWASP ZAP to detect common vulnerabilities and misconfigurations.

4. Manual Testing

Tested input fields and authentication mechanisms for unusual behavior.

Vulnerability 1

Cross-Site Scripting (XSS)

Risk Level: Medium (Moderate Security Risk)

Description:

The application allows user input that may not be properly sanitized, enabling injection of malicious scripts.

Impact:

Attackers can execute scripts in the user's browser, potentially stealing session data or performing unauthorized actions.

Recommendation:

Apply input validation and output encoding to prevent script injection.

Vulnerability 2

Security Misconfiguration

Risk Level: Medium

Description:

The application exposes certain system details and lacks strict security configurations.

Impact:

Attackers can gather useful information about the system and exploit weaknesses.

Recommendation:

Disable unnecessary features and implement secure configuration practices.

Vulnerability 3

Weak Authentication

Risk Level: High (Critical Risk)

Description:

The authentication mechanism may allow weak passwords or lacks strong validation controls.

Impact:

Unauthorized users may gain access to accounts, leading to data compromise.

Recommendation:

Enforce strong password policies and implement secure authentication mechanisms.

Evidence

The following evidence was collected during the assessment:

- Nmap scan results showing network information
 - OWASP ZAP alerts identifying vulnerabilities
 - Screenshots from manual testing

These results support the identified security findings.

Conclusion

The assessment identified several vulnerabilities within the application, including XSS, weak authentication, and security misconfigurations.

This highlights the importance of implementing secure coding practices and proper security controls to protect users and systems.

Learning Outcome

- Gained practical experience using Nmap and OWASP ZAP
- Understood common web vulnerabilities and their impact
- Improved skills in analyzing and documenting security findings
- Learned how real-world vulnerability assessments are performed